YuJia Liang

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Education

National Taiwan Ocean University — Keelung, Taiwan
B.S. in Computer Science and Engineering — *Sep. 2022 – Jun. 2026*
Graduate-Level Coursework (9 courses, 27 credits): Machine Learning, Generative AI, Natural Language Processing, Data Mining, Information Retrieval, Advanced Database Systems, Formal Methods, Smart IoT Platform

Relevant Courses: Data Structures & Algorithms, Graph Theory Algorithms, Operating Systems, Computer Organization, Computer Networks, Verilog HDL, Digital Logic, Linear Algebra, Probability Theory, Discrete Mathematics

Awards & Honors

Apple WWDC26 Swift Student Challenge — Winner 2026
One of 8 winners in Taiwan and 350 worldwide; covered by CNA, UDN, Liberty Times, ETtoday, and TechNews

NTOU CSE Capstone Competition — Silver Award & Honorable Mention Dec. 2025
Volleyball Match Analysis System (Silver in the Software Development Special Award) — led a 3-person team

Experience

Teaching Assistant — Introduction to iOS Application Development Sep. 2026 – Jan. 2027
Dept. of Computer Science and Engineering, National Taiwan Ocean University — *Keelung, Taiwan*

ITRI (Industrial Technology Research Institute) Jun. 2025 – Aug. 2025
AI Image Recognition Intern, Electronics and Optoelectronics Research Laboratories — *Hsinchu, Taiwan*

- Trained **YOLOPX**, a multi-task panoptic driving perception network performing object detection, drivable-area segmentation, and lane segmentation in a single forward pass; reached **0.972 mIoU** (drivable area), **0.793 mIoU** (lane), and **0.838 mAP@0.5** over 4 classes after 200 epochs on ITRI's in-house driving dataset
- Extended the open-source YOLOPX codebase by **∼1,900 lines** (6 files rewritten, 5 added) for ITRI's data format, building a YOLO-segmentation loader and a Mosaic/MixUp/Albumentations augmentation pipeline to replace one that handled BDD100K only
- Owned the **edge deployment path**: applied network pruning, exported to ONNX, wrote a tool resolving dynamic convolution-bias subgraphs that blocked TensorRT conversion, built a **TensorRT/PyCUDA engine for NVIDIA Jetson Nano**, and streamed images and detections from it to a desktop PC over **Ethernet/ROS at 30 FPS with sub-50ms latency**
- Delivered a command-line **batch-inference tool for ARTC** (Automotive Research & Testing Center) running the trained model across image and video datasets including a new rainy-weather set; separately reproduced and benchmarked **YOLOP, YOLOPv2, YOLOPX, and A-YOLOM** to identify deployment-viable designs, presenting findings at weekly research reviews

Research

ToneLeak — Counterfactual Auditing of Guitar Tone Models 2026
Python, PyTorch, scikit-learn, librosa, NumPy — *Independent Research*

- Testing whether guitar-tone models alter performance attributes that should stay attributable to the player (bends, vibrato, attack, dynamics), via a **paired counterfactual design** where a deterministic effect on the identical DI performance supplies ground truth
- Audited **2,040 published 64-dim Fx embeddings** over 680 renders of the public Open-Amp encoder: tone decodes at **99.3% balanced accuracy** but source combination stays at **87.3%** — tone-informative yet **not source-invariant**; a held-out causal reference swap (480 swaps over 120 query–tone clusters) shifted output RMS envelopes by **3.83 dB** (95% CI 3.28–4.39) under an identical query DI and intended tone
- Reported a **negative result**: projecting out 5 source-predictive dimensions cut source decodability from 82.6% to 47.3% but raised same-DI target error 53–78%, so it is not a viable mitigation; grounded the claim in a documented novelty audit against ISMIR/DAFx/ICASSP prior work

Projects

Single-Camera Volleyball Analysis – Capstone, continued as Research Nov. 2024 – Present
Python, PyTorch, YOLO, VballNet, BoT-SORT, OpenCV, FastAPI, React, Docker *Advisor: Prof. Pei-Yi Ding*

- Led a 3-person capstone team (50% of the work) combining VballNet ball tracking and in/out landing judgment with a **YOLOv11m** detector for 5 actions, trained on 24,806 images merged from two public datasets: **validation mAP@0.5 0.945** (precision/recall 0.916/0.915) — **Silver Award** and Honorable Mention in the department capstone competition
- Continued it solo as research, rebuilding the pipeline into versioned, cached stages that report why they could not produce a result: court registration that follows a panning, zooming camera (**0.10 m** median court-position error on held-out broadcast clips) and player tracking with a fine-tuned YOLO26s + BoT-SORT (**IDF1 0.690, HOTA 0.636** on SportsMOT volleyball)
- Replaced the capstone detector with a per-player action classifier weighted by the ball's position (macro F1 **0.303** → **0.678** on 16 unseen matches), added a rally classifier on the system's own cues (**F1 0.742** vs 0.423 for a ball-only rule) and ballistic 3D ball flights anchored to the touching player
- Built a React + FastAPI match review app with a timeline, 2D/3D tactics boards and per-player statistics; every reported number is produced by a script on a labelled split

RiffNode – AI Guitar Effects Studio for iPadOS & macOS 2026
Swift 6, SwiftUI, AVFoundation, Accelerate/vDSP, Foundation Models *WWDC26 Swift Student Challenge*

- Built a guitar effects studio entirely from native Apple frameworks, structured so business logic never depends on the UI or hardware code around it — keeping the audio engine independently testable and reusable — **WWDC26 Swift Student Challenge Winner**
- Engineered a low-latency `AVAudioEngine` pipeline driving 11 effects in a drag-and-drop chain under **Swift 6 strict concurrency**; keeping every pedal wired to the graph and treating enabled state as a bypass flag makes toggles glitch-free and confines rebuilds to reordering, while Accelerate/vDSP FFT and autocorrelation pitch detection drive a tuner, metronome, and chord detection
- Rebuilt the assistant as a **tool-calling agent** over 13 rig tools with a multi-turn observe-act-report loop and undo, clamping every model-proposed value to its `ParameterSpec` range so no model output reaches the DSP unchecked; one inference protocol backs **5 interchangeable providers** (on-device Apple Foundation Models by default and fully offline, Claude Code CLI, Codex CLI, Anthropic API, keyword fallback) with retrieval over an on-device NaturalLanguage embedding corpus

Technical Skills

Programming: Python, C++, Rust, Swift, TypeScript/JavaScript, Dart, SQL
ML & Computer Vision: PyTorch, TensorFlow, OpenCV, ONNX Runtime, TensorRT, ROS, Jetson Nano, model pruning & quantization, RAG & embedding retrieval, scikit-learn, NumPy, librosa
Apple Platforms: SwiftUI, AVFoundation, Accelerate/vDSP, ScreenCaptureKit, Core Image, NaturalLanguage, Foundation Models
Web & Backend: React, Next.js, Tailwind CSS, TypeScript, FastAPI, Celery, Flutter
Data & Infrastructure: Docker, GitHub Actions, GCP, Git, Redis, PostgreSQL, MongoDB, Firebase, pytest, SwiftPM
Languages: Mandarin (Native), English (Fluent, TOEFL 110, TOEIC 920), Japanese (JLPT N3)

Leadership & Activities

Volleyball Department Team Captain 2022 – present
Led the team to the top 8 at the national Information Departments Cup *Keelung, Taiwan*

NTUFC Kronos Quant Club — Team Pyxis Sep. – Dec. 2025
3 CTA alphas on OKX perpetuals — 283% in a 22-month backtest (Sharpe 4.76), deployed on GCP

NTU Photography Club (NTUPhoto) 2025 – present
Member & Darkroom Officer *Taipei, Taiwan*

Volunteer — Chengdu Research Base of Giant Panda Breeding Aug. 2024
16-hour giant panda conservation practicum *Chengdu, China*

PADI Open Water Diver (2023) · **10 km Kayak to Keelung Islet** (2026)
Certified scuba diver; completed NTOU's graduate kayak challenge

Hackathon Participant 2022 – present
Hotai Motor and TSMC corporate hackathons